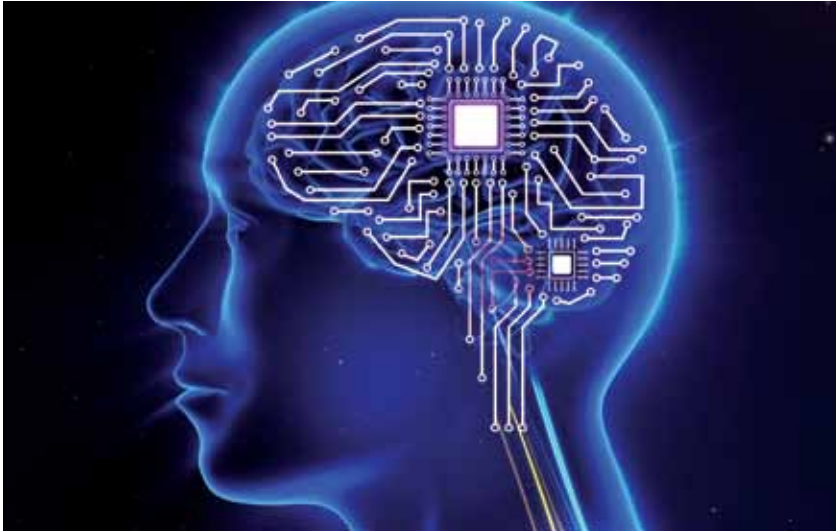


Open source software is helping the development of AI



Over the past few years, the importance of open source software in the realm of AI has increased significantly. Open source software has several advantages, one of which is the possibility for programmers to work together and exchange information. AI developers can build on the work of others and share their own contributions by adopting open source software, which promotes speedy innovation and growth in the field of AI.

The availability of numerous tools and resources for academicians and developers is another advantage of open source AI software. Many AI applications call for specialised tools and resources that may not be accessible to or be expensive for individuals or small businesses. These tools can be accessed and used without restriction when using open source software, which can help level the playing field and enable smaller businesses to compete with bigger ones.

Fortuna, a toolbox for assessing uncertainty in an ML model, is now available

Fortuna, an open source toolbox for machine learning (ML) model uncertainty assessment, has been made available, according to an announcement from AWS. Any trained neural network can generate calibrated uncertainty estimates using the calibration techniques provided by Fortuna, such as conformal prediction.

There are several published ways for calculating or calibrating the uncertainty of predictions, but the present tools and libraries for doing so have a narrow scope and do not offer a wide variety of methods. Fortuna fills this gap by combining well-known strategies and making them accessible to users through a standardised and user-friendly interface.

Fortuna has three usage modes, the first one being ‘starting from uncertainty estimates’. This mode is the quickest level of interaction with the library and has the fewest compatibility requirements. This usage mode provides conformal prediction techniques for both regression and classification.

PaLM + RLHF, a ChatGPT alternative that is not quite ready yet

The first open source ChatGPT equivalent is out. It is an application of RLHF (Reinforcement Learning with Human Feedback) built on top of Google’s PaLM architecture, which includes 540 billion parameters. It functions like ChatGPT but is still a work-in-progress. It combines PaLM, a sizable language model from Google, with RLHF to build a system that can practically execute every function that ChatGPT can, including email authoring and code recommendations.

PaLM + RLHF is not pre-trained. In order for the system to actually work, it must have acquired the crucial training utilising sample data from the web. After downloading PaLM + RLHF, a ChatGPT-like experience won’t happen suddenly; that would require producing gigabytes of text from which the model can learn and finding hardware able to handle the training demand. PaLM + RLHF won’t be able to take the place of ChatGPT unless a well-funded initiative (or individual) goes to the trouble of educating and making it available to the general public.



Both ChatGPT and PaLM + RLHF share a secret sauce called Reinforcement Learning with Human Feedback, which aims to better match language models with what users want them to do. In RLHF, a language model is fine-tuned using a data set that includes prompts that correspond to what volunteers’ expectations for the model are, such as “Machine learning is a form of AI...” The language model used in PaLM + RLHF is PaLM.